QuChaTeR: A Hybrid Quantum-Chaotic Temporal Framework for Earthquake Prediction

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Introduction

- Signal processing is required to try to forecast earthquakes, and this is mostly done with classical algorithms and computers.

- This study examines a framework that makes earthquake predictions using a hybrid algorithm (combination of quantum and classical algorithms).

Motivation

- Earthquake signals are nonlinear, chaotic, and non-stationary.

- Classical models fail to capture complex temporal dynamics.

- Can chaos theory + quantum computing improve prediction?

Related Works

- **Chaos-based modeling:**
  Yi et al. (2010) → Captures nonlinear/chaotic patterns in seismic time series
- **Wavelet-based methods:**
  Yamamoto & Baker → Multi-scale temporal feature extraction
- **Deep learning models:**
  Hamdi et al. (2024) → Bi-LSTM outperforms LSTM (long-term dependencies)
  Sneka & Kanchana (2025) → CNN–LSTM improves geolocation-based forecasting
- **Quantum-inspired models:**
  Dutta et al. (2025) → Hybrid quantum neural networks for tsunami prediction

[1] Yi et al., ICISS, 2010
[2] Yamamoto & Baker, BSSA, 2013
[3] Hamdi et al., IJECBE, 2024
[4] Sneka & Kanchana, JAICN, 2025
[5] Dutta et al., EPJ QT, 2025

Dataset & Preprocessing

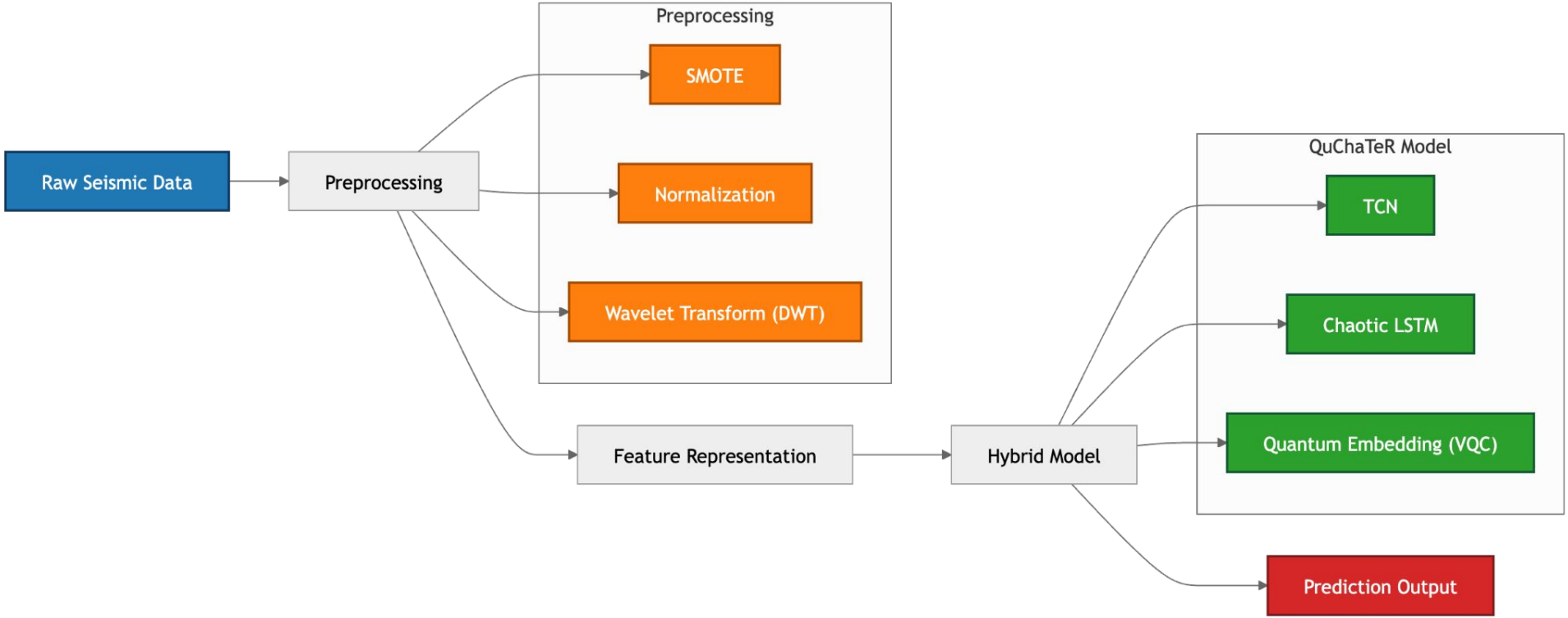
- **Dataset:** Northern California Earthquake Data Center — 512-hour seismic time-series used for binary earthquake prediction.

Preprocessing

- Class imbalance handled using SMOTE.
- Signals normalized and missing values imputed.
- Wavelet Transform (DWT) applied for noise reduction and multi-scale feature extraction.

[6] https://www.timeseriesclassification.com/description.php?Dataset=Earthquakes

QuChaTeR Architecture Overview



QuChaTeR in One Equation

$$\mathbf{h}_t = \mathcal{Q}\Big(\mathcal{H}\big(r \cdot \sigma(\mathrm{TCN}(\mathbf{x}_t)) \odot \big[1 - \sigma(\mathrm{TCN}(\mathbf{x}_t))\big]\big)\Big)$$

The Chaotic Engine: Nonlinear Perturbation

Chaotic Injection

- Logistic Map perturbation: $\mathbf{z}_t = r\,\mathbf{h}_t \odot (1 - \mathbf{h}_t)$
- Hénon Map dynamics: $x_{t+1} = 1 - ax_t^2 + y_t, \quad y_{t+1} = bx_t$

Parameters

- $a = 1.4,\ b = 0.3$ (Bounded chaotic regime)
- $r^* = 3.8475$ (Bayesian optimization result)

[7] May, Nature, 1976
[8] Hénon, CMP, 1976

Quantum Embedding Layer

$$U(\mathbf{x}_t, \Theta)$$

$$\mathbf{q}_t = [\langle Z_1 \rangle, \ldots, \langle Z_Q \rangle]^\top$$

$$\mathbf{h}_t^{(q)} = \mathbf{h}_{t-1}^{(q)} + W_q \mathbf{q}_t$$

Configuration

- 6-qubit variational circuit
- Rotation gates (R_Y, R_Z)+ entanglement (CNOT)

Experimental Setup

Models Compared

- RNN, LSTM, GRU, 1D-CNN
- Reservoir Computing
- Quantum LSTM

Training Setup

- 50 epochs, Adam optimizer (lr = 0.001)
- Same conditions for all models (fair comparison)
- Train / validation / independent test split
- NVIDIA L4 GPU

Quantum Setup

- PennyLane simulator (noise-free)
- 6-qubit configuration selected

Evaluation Metrics

- Accuracy, Precision, Recall, F1-score
- Generalization Stability (G) metric

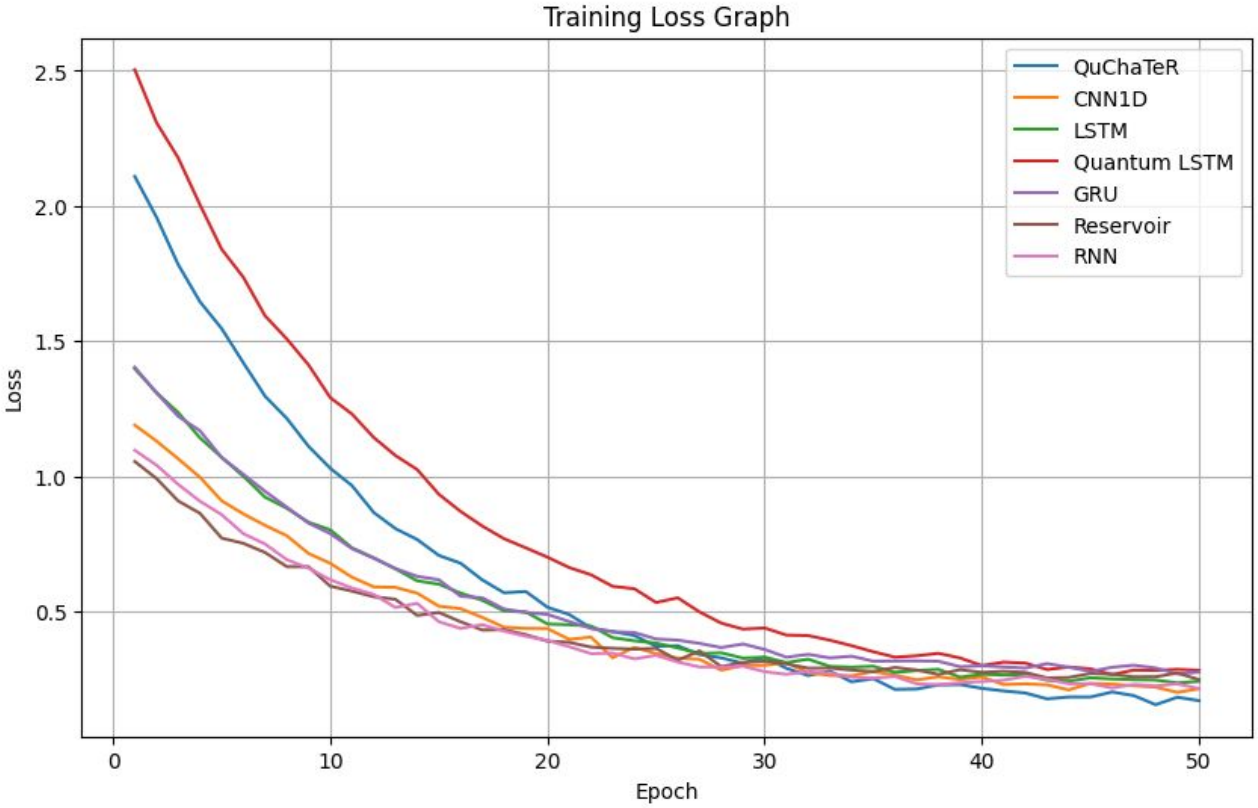
$$G = A_{test} - |A_{train} - A_{test}|$$

Results

Model	Accuracy	Recall	Precision	F1	ROC-AUC
QuChaTeR	0.9634	0.9682	0.9501	0.9590	0.9785
CNN1D	0.9215	0.9083	0.9447	0.9258	0.9407
LSTM	0.9057	0.8815	0.9123	0.8969	0.9225
Quantum LSTM	0.8931	0.9052	0.8745	0.8891	0.9139
GRU	0.8553	0.8415	0.8624	0.8519	0.8829
Reservoir	0.8335	0.8188	0.8351	0.8266	0.8635
RNN	0.7918	0.7710	0.8035	0.7869	0.8323

Qubits	Train Accuracy	Test Accuracy	$\lvert Train - Test \rvert$	G
2	0.8813	0.8542	0.027	0.827
4	0.9612	0.8807	0.081	0.800
6	0.9902	0.9634	0.027	**0.937**
8	0.9527	0.9141	0.039	0.876

Results


Training Loss Graph
QuChaTeR
CNN1D
LSTM
Quantum LSTM
GRU
Reservoir
RNN
Loss
2.5
2.0
1.5
1.0
0.5
0
10
20
30
40
50
Epoch

Discussion

- This study has shown that quantum-hybrid architectures are quite successful.

- However, it should be noted that this study is not yet possible with current quantum computers. Therefore, it was carried out using simulators.

Future Work

- Evaluation on real quantum hardware (NISQ devices)
- Scaling to larger and more diverse seismic datasets
- Exploration of deeper quantum circuits and architectures

Thank You for Your Attention

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Codes: https://github.com/emirkaanozdemr/QuChaTeR

Questions?